

When a Basis Goes Stale: Routing Consequence Through Admissibility

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Abstract

When a governance basis becomes stale, the question facing any consequence-bearing system is what form of continuation is permissible. Binary suspend-or-proceed models collapse a multi-dimensional decision into a single bit, discarding information that determines whether continuation is ungoverned, admissibility-controlled, or structurally prohibited. This paper specifies a routing matrix for consequence resolution after admissibility failure, grounded in a six-state continuation taxonomy (pass, contain, refuse, suspend, escalate, require re-attestation) with `HARD_STOP` as a terminal veto, keyed by a base matrix of failure kind against consequence grade, and modified by authority state, reversibility, escalation availability, and policy reference. The specification is drawn from the Aurora-Lens runtime epistemic governance architecture, where the base matrix and modifier pipeline are implemented and tested. Two extension points — absent ownership as a canonical failure kind, and compound staleness composition rules — are named as open and given intended design direction without overclaiming implementation status.

1. The Problem a Binary Outcome Cannot Solve

Consider a diagnostics vendor acquired by a larger conglomerate. A hospital system that shares patient data with the vendor under a signed delegation does what every well-run compliance function does after an acquisition: legal confirms the new entity, the signing authority is re-registered, the delegation is re-attested. Ownership of the relationship transfers without a gap.

Nobody re-examines whether the original clinical rationale for the data-sharing scope — a specific diagnostic model class, a defined retention window, a consent boundary drawn under the old company's data-handling practices — still holds under the new parent's. The re-attestation already happened. The box is ticked. Ownership is current, so the whole delegation reads as current, because the form only asked one question and the acquisition answered it convincingly.

The delegation is running on a stale legitimacy basis. The ownership signature is fresh. The operational-legitimacy signature has silently expired. A binary suspend-or-proceed model applied at this moment faces a question it cannot answer with the vocabulary it has. The delegation is neither fully valid nor fully invalid. Its ownership basis is current. Its legitimacy basis is stale. The consequence tier is high — patient data, regulated retention, a consent boundary. The action is irreversible — once shared, the data cannot be unshared.

A system that can say only “stop” or “go” will either halt a delegation whose ownership is perfectly current, or permit continuation on a legitimacy basis that has expired. Both are wrong. The first is operationally destructive. The second is governancely negligent. The information needed to choose correctly — which basis failed, at what consequence tier, with what reversibility profile, and what escalation paths exist — is present in the system. A binary outcome model discards it.

2. The Continuation Taxonomy and Terminal Veto

The Aurora-Lens architecture resolves admissibility evaluations into six continuation states and one terminal veto. Each represents a structurally distinct form of governed response, and the distinctions between them are load-bearing.

PASS. The basis is current. Execution continues under full admissibility. No governance intervention is required. PASS is included here for completeness as the non-failure baseline; it is not a stale-basis outcome.

CONTAIN. The basis has degraded or become incomplete in a way that prevents ordinary pass-through but does not yet require removal of all consequence-bearing authority. Execution continues under monitoring constraints that narrow the permissible action space. The system is on notice. Contain is the governance equivalent of a warning corridor: the action proceeds, but the next evaluation will apply tighter criteria.

REFUSE. The specific action is inadmissible, but the broader execution context remains governed. The system declines this particular consequence while preserving the ability to evaluate subsequent actions. Refuse is action-scoped, not session-scoped.

SUSPEND. Consequence-bearing authority is removed until the basis is restored. Execution does not continue in any form that carries consequence. Suspend is the appropriate outcome when the failure is serious enough that no form of monitored continuation is defensible, but the failure is recoverable — the basis can, in principle, be re-established.

ESCALATE. The failure requires adjudication by a named authority that the current evaluation layer cannot provide. Execution routing transfers upward. Escalate is only available when an escalation path exists and has been pre-configured; it is never a default, and its absence forces fallback to a more restrictive outcome.

REQUIRE RE-ATTESTATION. The basis has not yet failed, but its continued validity depends on an assertion that has not been renewed within the required window. The system holds consequence until the attestation is provided. This is distinct from suspend: the basis is not known to be invalid, but it is no longer known to be valid, and the difference between those two states is governable.

HARD_STOP. Consequence is categorically prohibited. Unlike SUSPEND, which permits restoration after re-establishing the basis, HARD_STOP closes the present commitment path and requires a new lawful initiation before consequence can be attempted again. HARD_STOP is a terminal veto, reserved for conditions where continued execution on the current basis is impermissible regardless of what remediation is offered within the existing path.

These six continuation states describe the governed continuation space. HARD_STOP operates outside that space as a terminal veto. Any admissibility evaluation resolves to exactly one of these seven outcomes. The routing logic that determines which outcome applies to a given failure is the subject of the next three sections.

3. The Base Matrix: Failure Kind × Consequence Grade

The routing matrix operates in two passes. The first pass establishes a base outcome by crossing the kind of admissibility failure against the consequence grade of the action being attempted.

3.1 Failure Kinds

Three canonical failure kinds are currently defined:

Stale evidence. The evidentiary basis for the action — a fact, a status, a score, a certification — was valid at the time of original evaluation but has not been re-validated within the required freshness window. The supplier score set at onboarding and never rechecked is a stale-evidence failure.

Missing freshness. The system expected a freshness signal — a timestamp, a re-validation event, a currency attestation — and none was received. This is distinct from stale evidence: stale evidence had a valid basis that aged out; missing freshness never received the signal that would establish currency in the first place. Absence of an expected freshness signal is itself an admissibility condition, not an information gap to be filled by assumption.

Unvalidated orientation. The external orientation state — a sanctions status, a regulatory classification, a policy version, a role authority — has changed or may have changed, and the action's admissibility basis has not been re-evaluated against the current orientation. This failure kind was introduced to handle the class of cases where the world moves and the governance basis does not move with it. Silent decay — continued execution under conditions that have shifted since the basis was established — is the characteristic failure mode.

3.2 Consequence Grades

Five consequence grades are defined, each representing a tier of impact if the action proceeds on an inadmissible basis:

- **Critical:** Irreversible harm, regulatory breach, safety-of-life, or fundamental rights violation.
- **High:** Significant institutional, financial, legal, or reputational consequence. The hospital data-sharing case sits here.
- **Moderate:** Material operational consequence with recovery pathways.
- **Low:** Minor operational consequence, contained impact radius.
- **Minimal:** Negligible consequence; failure affects record-keeping more than outcomes.

3.3 The Base Matrix

The base matrix establishes the starting outcome for each failure-kind × consequence-grade intersection. The governing principle is: **consequence grade sets the floor, and the floor rises monotonically with the grade.**

	Minimal	Low	Moderate	High	Critical
Stale evidence	CONTAIN	CONTAIN	ESCALATE	SUSPEND	HARD_STOP
Missing freshness	CONTAIN	ESCALATE	SUSPEND	SUSPEND	HARD_STOP
Unvalidated orientation	RE-ATTEST	RE-ATTEST	ESCALATE	SUSPEND	HARD_STOP

Three structural properties of this matrix:

First, **critical consequence always resolves to HARD_STOP regardless of failure kind.** This is an absolute veto. A system that permits consequence-bearing execution at the critical tier on any form of stale, missing, or unvalidated basis has made a design error, not a design choice.

Second, **missing freshness is treated more severely than stale evidence at the same consequence grade**. Stale evidence at least had a valid basis once; missing freshness never established one. The absence of an expected signal is a stronger admissibility failure than the aging of a received one.

Third, **unvalidated orientation routes to RE-ATTEST at lower tiers rather than CONTAIN**. The rationale: when the external orientation has shifted, monitoring alone (CONTAIN) is insufficient because the monitoring criteria themselves may be calibrated against the old orientation. Re-attestation forces the evaluation to update against the current state before consequence resumes.

4. The Modifier Pipeline

The base matrix establishes a starting outcome. The modifier pipeline can tighten or redirect that outcome — but never relax it. The pipeline operates in the following order:

4.1 Authority State

If the authority state of the signer or governing entity is **absent** (not merely stale, but structurally gone — the role is vacant, the entity no longer exists, the signer has been de-authorised), the outcome is forced to SUSPEND or HARD_STOP regardless of the base matrix result. Authority absence overrides all other modifiers.

If the authority state is **contested** (two competing claims, an unresolved succession, an ambiguous delegation), the outcome is tightened by at least one severity step from the base matrix result. A base CONTAIN becomes ESCALATE. A base ESCALATE becomes SUSPEND.

If the authority state is **current**, no modification is applied.

4.2 Escalation Availability

ESCALATE is only a valid outcome when a named escalation authority exists and has been pre-configured for this failure class. If the base matrix or a prior modifier produces ESCALATE and no escalation path is available, the outcome falls back deterministically:

- At high or critical consequence: falls to SUSPEND or HARD_STOP.
- At moderate consequence: falls to SUSPEND.
- At low or minimal consequence: falls to RE-ATTEST.

The absence of an escalation path is itself a governance signal. A system that defaults to ESCALATE without a configured recipient is routing consequence into a void.

4.3 Reversibility

Reversibility does not relax outcomes. A reversible action at a given consequence grade receives the same base matrix outcome as an irreversible one — because admissibility is a property of the basis, not of the recoverability of the action.

Where reversibility modifies behaviour is in the *form* of a CONTAIN or RE-ATTEST outcome: a reversible action under CONTAIN may proceed with a tighter monitoring window; an irreversible action under CONTAIN may be tightened to RE-ATTEST. The governing principle: reversibility affects how containment operates, not whether containment is sufficient.

4.4 Policy Reference

The presence of an explicit policy reference — a governance rule, a compliance requirement, a regulatory citation — activates a warning corridor within CONTAIN outcomes. A CONTAIN result with a policy reference generates an auditable warning linked to the specific policy, creating a traceable record that the system identified a controlled deviation and continued under stated governance authority. A CONTAIN result without a policy reference is a weaker claim: the system continued, but the basis for continuation is architectural default rather than stated policy.

5. Extension Points

Two areas of the routing architecture are specified as extension points — designed for, not yet canonically implemented.

5.1 Absent Ownership as a Canonical Failure Kind

The three current failure kinds (stale evidence, missing freshness, unvalidated orientation) all describe failures of informational or evaluative currency. Absent ownership — the condition where the accountable entity for a delegation has ceased to exist, departed, or been structurally removed — is a failure of a different category: accountability itself.

In the current architecture, absent ownership is governed through trust and authority pathways rather than the freshness/revalidation policy. It is a designated extension point for canonicalisation as a fourth failure kind. The intended treatment: absent ownership resolves as an absolute veto (SUSPEND or HARD_STOP) unless governance policy explicitly attributes authority to the role rather than the individual — meaning the title carries the authority, and succession into the title inherits it. Where role-binding is configured, absent ownership is re-evaluated as a trust-continuity question rather than an accountability-absence question.

5.2 Compound Staleness: Multi-Kind Composition

The current architecture evaluates one failure kind at a time. It does not yet model compound states — stale evidence co-occurring with unvalidated orientation, or missing freshness co-occurring with absent authority. The composition semantics for simultaneous multi-kind failures are an open design space.

The intended direction is **conservative dominance composition**:

- If any participating failure kind would independently resolve to HARD_STOP, the compound resolves to HARD_STOP.
- Otherwise, the compound resolves to the strictest permissible outcome across the participating kinds, after authority and escalation constraints have been applied.
- Compound failures at moderate consequence or above are treated as at least SUSPEND, on the principle that simultaneous failures in independent dimensions constitute stronger evidence of basis degradation than either failure alone.

For example: stale evidence (base: CONTAIN at low consequence) co-occurring with unvalidated orientation (base: RE-ATTEST at low consequence) would resolve under conservative dominance to RE-ATTEST — the stricter of the two — with an auditable record that both kinds contributed to the outcome.

This composition rule is specified as intended design direction. It is not yet implemented as wired policy.

6. The Distinction That Governs

The hospital's delegation ran on a stale legitimacy basis. The ownership signature was fresh. The operational-legitimacy signature had silently expired. Under the routing architecture specified here, that failure is classifiable: an unvalidated-orientation failure at high consequence grade, producing a base outcome of SUSPEND, with no modifier relaxing it. The delegation stops. Consequence-bearing authority is removed until the legitimacy basis is re-established against current conditions.

Under a binary model, the same failure produces either STOP (operationally destructive — the ownership is fine) or PROCEED (governance negligent — the legitimacy is stale). Under the continuation taxonomy, it produces SUSPEND: a structurally distinct outcome that says the delegation's ownership is not in question, but its legitimacy basis must be re-validated before consequence resumes. That distinction preserves the operational relationship while removing the governance failure.

The deeper claim is this: a delegation running on a stale legitimacy basis that nobody has flagged is ungoverned continuation wearing the paperwork of a governed one. The routing matrix specified here is the mechanism that makes the difference between ungoverned and admissibility-controlled continuation concrete, auditable, and enforceable. Admissibility is earned. When the basis degrades, what follows is not a binary — it is a routing decision through six continuation states and a terminal veto, determined by what failed, at what consequence, with what authority, and with what escalation paths available.

Capability creates value. Accountability, proven at the moment of action, determines deployability. A system that cannot distinguish between a fresh basis and a stale one wearing current paperwork has not implemented governance. It has implemented the appearance of governance, and the appearance will hold until the first silent failure reaches consequence.

Acknowledgments

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The Aurora-Lens architecture, from which this specification is drawn, is the subject of Australian provisional patent applications 2025905835, 2025905860, 2025905885, 2025906132, 2025906680, and 2026905969, and PCT application PCT/AU2026/050569.

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